Model Architecture: Llama-3.2-1B

Generated by sg_modelload

July 21, 2026

1 Model Overview

Model Name	Llama-3.2-1B
Path	/models/memorization/Llama-3.2-1B
Total Parameters	1,235,814,400
Weight Dtype	bfloat16

2 Forward Pass Algorithm

Algorithm 1 Forward pass of Llama-3.2-1B

$h \leftarrow \text{embed_tokens}(x) \in \mathbb{R}^{B \times T \times 2048}$
for $layer \leftarrow 0$ **to** 15 **do**
    $h \leftarrow \text{layer}_{layer}(h)$
end for
$h \leftarrow \text{norm}(h) \in \mathbb{R}^{B \times T \times 2048}$
$\text{logits} \leftarrow \text{lm_head}(h) \in \mathbb{R}^{B \times T \times 128256}$

3 Layer-by-Layer Architecture

All weight tensors and their dimensions are listed below.

Layer Path	Type	Weight Shape	Dtype	Params
model.layers.0	LlamaDecoderLayer	–	–	–
model.layers.0.input_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.0.mlp	LlamaMLP	–	–	–
model.layers.0.mlp.down_proj.weight	Linear	2048 × 8192	bfloat16	16,777,216
model.layers.0.mlp.gate_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.0.mlp.up_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.0.post_attention_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.0.self_attn	LlamaAttention	–	–	–
model.layers.0.self_attn.k_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.0.self_attn.o_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.0.self_attn.q_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.0.self_attn.v_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.1	LlamaDecoderLayer	–	–	–
model.layers.1.input_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.1.mlp	LlamaMLP	–	–	–
model.layers.1.mlp.down_proj.weight	Linear	2048 × 8192	bfloat16	16,777,216
model.layers.1.mlp.gate_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.1.mlp.up_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.1.post_attention_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.1.self_attn	LlamaAttention	–	–	–
model.layers.1.self_attn.k_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.1.self_attn.o_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.1.self_attn.q_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304

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Layer Path	Type	Weight Shape	Dtype	Params
model.layers.1.self_attn.v_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.2	LlamaDecoderLayer	–	–	–
model.layers.2.input_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.2.mlp	LlamaMLP	–	–	–
model.layers.2.mlp.down_proj.weight	Linear	2048 × 8192	bfloat16	16,777,216
model.layers.2.mlp.gate_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.2.mlp.up_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.2.post_attention_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.2.self_attn	LlamaAttention	–	–	–
model.layers.2.self_attn.k_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.2.self_attn.o_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.2.self_attn.q_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.2.self_attn.v_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.3	LlamaDecoderLayer	–	–	–
model.layers.3.input_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.3.mlp	LlamaMLP	–	–	–
model.layers.3.mlp.down_proj.weight	Linear	2048 × 8192	bfloat16	16,777,216
model.layers.3.mlp.gate_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.3.mlp.up_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.3.post_attention_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.3.self_attn	LlamaAttention	–	–	–
model.layers.3.self_attn.k_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.3.self_attn.o_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.3.self_attn.q_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.3.self_attn.v_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.4	LlamaDecoderLayer	–	–	–
model.layers.4.input_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.4.mlp	LlamaMLP	–	–	–
model.layers.4.mlp.down_proj.weight	Linear	2048 × 8192	bfloat16	16,777,216
model.layers.4.mlp.gate_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.4.mlp.up_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.4.post_attention_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.4.self_attn	LlamaAttention	–	–	–
model.layers.4.self_attn.k_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.4.self_attn.o_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.4.self_attn.q_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.4.self_attn.v_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.5	LlamaDecoderLayer	–	–	–
model.layers.5.input_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.5.mlp	LlamaMLP	–	–	–
model.layers.5.mlp.down_proj.weight	Linear	2048 × 8192	bfloat16	16,777,216
model.layers.5.mlp.gate_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.5.mlp.up_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.5.post_attention_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.5.self_attn	LlamaAttention	–	–	–
model.layers.5.self_attn.k_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.5.self_attn.o_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.5.self_attn.q_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.5.self_attn.v_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.6	LlamaDecoderLayer	–	–	–
model.layers.6.input_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.6.mlp	LlamaMLP	–	–	–
model.layers.6.mlp.down_proj.weight	Linear	2048 × 8192	bfloat16	16,777,216
model.layers.6.mlp.gate_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.6.mlp.up_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.6.post_attention_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.6.self_attn	LlamaAttention	–	–	–
model.layers.6.self_attn.k_proj.weight	Linear	512 × 2048	bfloat16	1,048,576

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Layer Path	Type	Weight Shape	Dtype	Params
model.layers.6.self_attn.o_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.6.self_attn.q_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.6.self_attn.v_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.7	LlamaDecoderLayer	–	–	–
model.layers.7.input_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.7.mlp	LlamaMLP	–	–	–
model.layers.7.mlp.down_proj.weight	Linear	2048 × 8192	bfloat16	16,777,216
model.layers.7.mlp.gate_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.7.mlp.up_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.7.post_attention_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.7.self_attn	LlamaAttention	–	–	–
model.layers.7.self_attn.k_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.7.self_attn.o_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.7.self_attn.q_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.7.self_attn.v_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.8	LlamaDecoderLayer	–	–	–
model.layers.8.input_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.8.mlp	LlamaMLP	–	–	–
model.layers.8.mlp.down_proj.weight	Linear	2048 × 8192	bfloat16	16,777,216
model.layers.8.mlp.gate_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.8.mlp.up_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.8.post_attention_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.8.self_attn	LlamaAttention	–	–	–
model.layers.8.self_attn.k_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.8.self_attn.o_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.8.self_attn.q_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.8.self_attn.v_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.9	LlamaDecoderLayer	–	–	–
model.layers.9.input_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.9.mlp	LlamaMLP	–	–	–
model.layers.9.mlp.down_proj.weight	Linear	2048 × 8192	bfloat16	16,777,216
model.layers.9.mlp.gate_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.9.mlp.up_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.9.post_attention_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.9.self_attn	LlamaAttention	–	–	–
model.layers.9.self_attn.k_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.9.self_attn.o_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.9.self_attn.q_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.9.self_attn.v_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.10	LlamaDecoderLayer	–	–	–
model.layers.10.input_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.10.mlp	LlamaMLP	–	–	–
model.layers.10.mlp.down_proj.weight	Linear	2048 × 8192	bfloat16	16,777,216
model.layers.10.mlp.gate_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.10.mlp.up_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.10.post_attention_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.10.self_attn	LlamaAttention	–	–	–
model.layers.10.self_attn.k_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.10.self_attn.o_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.10.self_attn.q_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.10.self_attn.v_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.11	LlamaDecoderLayer	–	–	–
model.layers.11.input_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.11.mlp	LlamaMLP	–	–	–
model.layers.11.mlp.down_proj.weight	Linear	2048 × 8192	bfloat16	16,777,216
model.layers.11.mlp.gate_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.11.mlp.up_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.11.post_attention_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048

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Layer Path	Type	Weight Shape	Dtype	Params
model.layers.11.self_attn	LlamaAttention	–	–	–
model.layers.11.self_attn.k_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.11.self_attn.o_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.11.self_attn.q_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.11.self_attn.v_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.12	LlamaDecoderLayer	–	–	–
model.layers.12.input_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.12.mlp	LlamaMLP	–	–	–
model.layers.12.mlp.down_proj.weight	Linear	2048 × 8192	bfloat16	16,777,216
model.layers.12.mlp.gate_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.12.mlp.up_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.12.post_attention_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.12.self_attn	LlamaAttention	–	–	–
model.layers.12.self_attn.k_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.12.self_attn.o_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.12.self_attn.q_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.12.self_attn.v_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.13	LlamaDecoderLayer	–	–	–
model.layers.13.input_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.13.mlp	LlamaMLP	–	–	–
model.layers.13.mlp.down_proj.weight	Linear	2048 × 8192	bfloat16	16,777,216
model.layers.13.mlp.gate_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.13.mlp.up_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.13.post_attention_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.13.self_attn	LlamaAttention	–	–	–
model.layers.13.self_attn.k_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.13.self_attn.o_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.13.self_attn.q_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.13.self_attn.v_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.14	LlamaDecoderLayer	–	–	–
model.layers.14.input_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.14.mlp	LlamaMLP	–	–	–
model.layers.14.mlp.down_proj.weight	Linear	2048 × 8192	bfloat16	16,777,216
model.layers.14.mlp.gate_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.14.mlp.up_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.14.post_attention_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.14.self_attn	LlamaAttention	–	–	–
model.layers.14.self_attn.k_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.14.self_attn.o_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.14.self_attn.q_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.14.self_attn.v_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.15	LlamaDecoderLayer	–	–	–
model.layers.15.input_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.15.mlp	LlamaMLP	–	–	–
model.layers.15.mlp.down_proj.weight	Linear	2048 × 8192	bfloat16	16,777,216
model.layers.15.mlp.gate_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.15.mlp.up_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.15.post_attention_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.15.self_attn	LlamaAttention	–	–	–
model.layers.15.self_attn.k_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.15.self_attn.o_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.15.self_attn.q_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.15.self_attn.v_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
lm_head.weight	Linear	128256 × 2048	bfloat16	262,668,288
model	LlamaModel	–	–	–
model.embed_tokens.weight	Embedding	128256 × 2048	bfloat16	262,668,288
model.layers	ModuleList	–	–	–
model.norm.weight	LlamaRMSNorm	2048	bfloat16	2,048

4 Parameter Count Estimation

Component	Parameters	% of Total
MLP	805,306,368	65.2%
Embedding	262,668,288	21.3%
LM Head	262,668,288	21.3%
Attention	167,772,160	13.6%
LayerNorm	67,584	0.0%
Total	**1,235,814,400**	**100.0%**

4.1 Per-Layer Parameter Count

Layer	Params	Running Total
model.layers.0.input_layernorm	2,048	2,048
model.layers.0.mlp.down_proj	16,777,216	16,779,264
model.layers.0.mlp.gate_proj	16,777,216	33,556,480
model.layers.0.mlp.up_proj	16,777,216	50,333,696
model.layers.0.post_attention_layernorm	2,048	50,335,744
model.layers.0.self_attn.k_proj	1,048,576	51,384,320
model.layers.0.self_attn.o_proj	4,194,304	55,578,624
model.layers.0.self_attn.q_proj	4,194,304	59,772,928
model.layers.0.self_attn.v_proj	1,048,576	60,821,504
model.layers.1.input_layernorm	2,048	60,823,552
model.layers.1.mlp.down_proj	16,777,216	77,600,768
model.layers.1.mlp.gate_proj	16,777,216	94,377,984
model.layers.1.mlp.up_proj	16,777,216	111,155,200
model.layers.1.post_attention_layernorm	2,048	111,157,248
model.layers.1.self_attn.k_proj	1,048,576	112,205,824
model.layers.1.self_attn.o_proj	4,194,304	116,400,128
model.layers.1.self_attn.q_proj	4,194,304	120,594,432
model.layers.1.self_attn.v_proj	1,048,576	121,643,008
model.layers.2.input_layernorm	2,048	121,645,056
model.layers.2.mlp.down_proj	16,777,216	138,422,272
model.layers.2.mlp.gate_proj	16,777,216	155,199,488
model.layers.2.mlp.up_proj	16,777,216	171,976,704
model.layers.2.post_attention_layernorm	2,048	171,978,752
model.layers.2.self_attn.k_proj	1,048,576	173,027,328
model.layers.2.self_attn.o_proj	4,194,304	177,221,632
model.layers.2.self_attn.q_proj	4,194,304	181,415,936
model.layers.2.self_attn.v_proj	1,048,576	182,464,512
model.layers.3.input_layernorm	2,048	182,466,560
model.layers.3.mlp.down_proj	16,777,216	199,243,776
model.layers.3.mlp.gate_proj	16,777,216	216,020,992
model.layers.3.mlp.up_proj	16,777,216	232,798,208
model.layers.3.post_attention_layernorm	2,048	232,800,256
model.layers.3.self_attn.k_proj	1,048,576	233,848,832
model.layers.3.self_attn.o_proj	4,194,304	238,043,136
model.layers.3.self_attn.q_proj	4,194,304	242,237,440
model.layers.3.self_attn.v_proj	1,048,576	243,286,016
model.layers.4.input_layernorm	2,048	243,288,064
model.layers.4.mlp.down_proj	16,777,216	260,065,280
model.layers.4.mlp.gate_proj	16,777,216	276,842,496
model.layers.4.mlp.up_proj	16,777,216	293,619,712
model.layers.4.post_attention_layernorm	2,048	293,621,760
model.layers.4.self_attn.k_proj	1,048,576	294,670,336
model.layers.4.self_attn.o_proj	4,194,304	298,864,640
model.layers.4.self_attn.q_proj	4,194,304	303,058,944
model.layers.4.self_attn.v_proj	1,048,576	304,107,520
model.layers.5.input_layernorm	2,048	304,109,568
model.layers.5.mlp.down_proj	16,777,216	320,886,784
model.layers.5.mlp.gate_proj	16,777,216	337,664,000
model.layers.5.mlp.up_proj	16,777,216	354,441,216
model.layers.5.post_attention_layernorm	2,048	354,443,264
model.layers.5.self_attn.k_proj	1,048,576	355,491,840
model.layers.5.self_attn.o_proj	4,194,304	359,686,144
model.layers.5.self_attn.q_proj	4,194,304	363,880,448
model.layers.5.self_attn.v_proj	1,048,576	364,929,024
model.layers.6.input_layernorm	2,048	364,931,072
model.layers.6.mlp.down_proj	16,777,216	381,708,288
model.layers.6.mlp.gate_proj	16,777,216	398,485,504
model.layers.6.mlp.up_proj	16,777,216	415,262,720
model.layers.6.post_attention_layernorm	2,048	415,264,768
model.layers.6.self_attn.k_proj	1,048,576	416,313,344
model.layers.6.self_attn.o_proj	4,194,304	420,507,648
model.layers.6.self_attn.q_proj	4,194,304	424,701,952